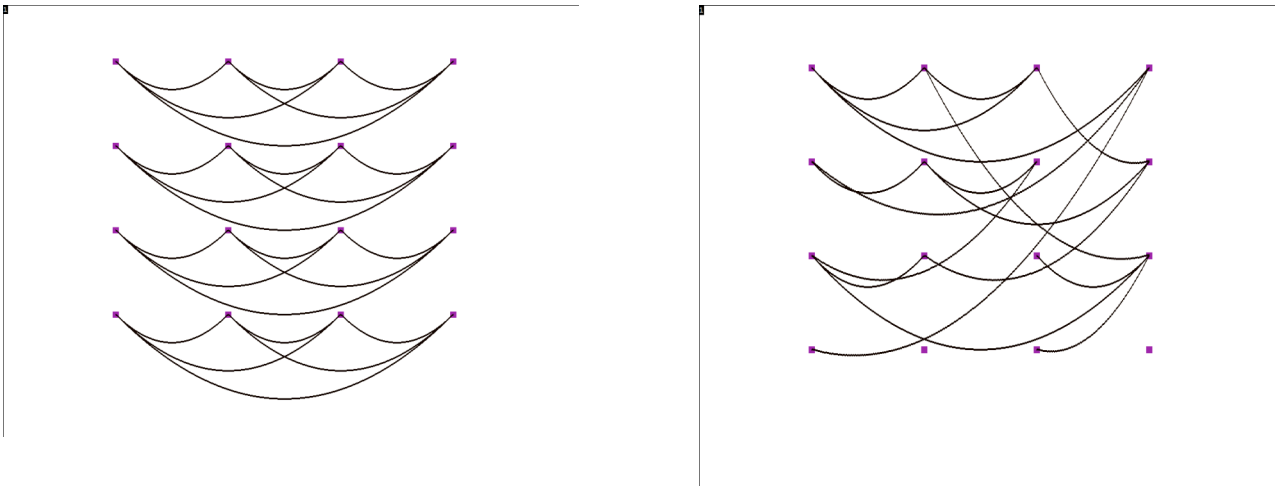
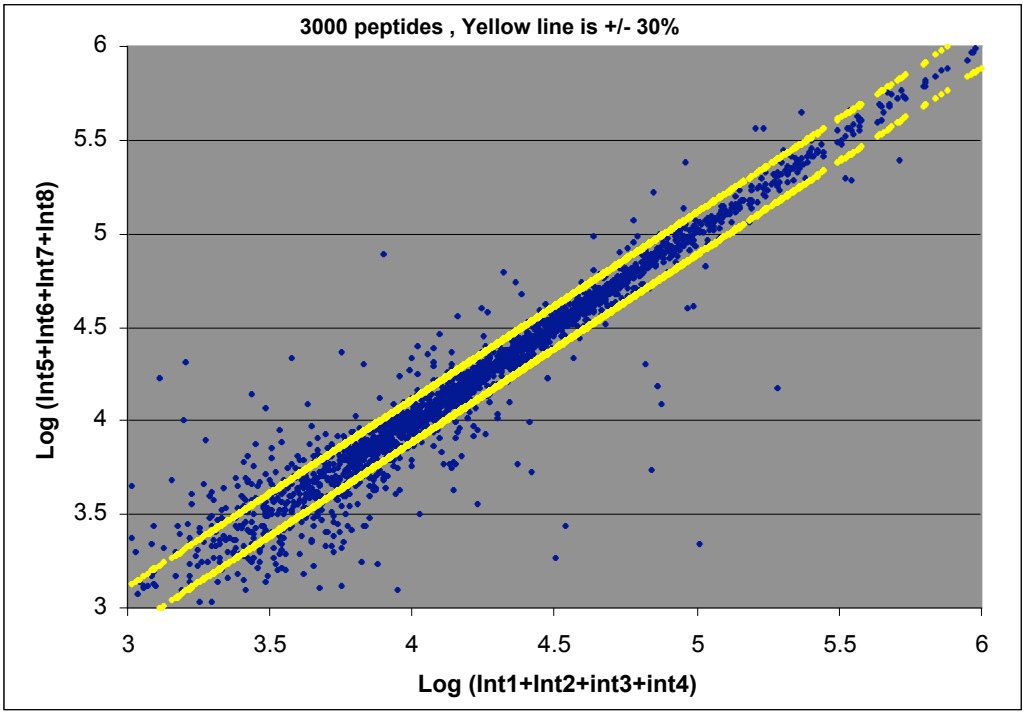
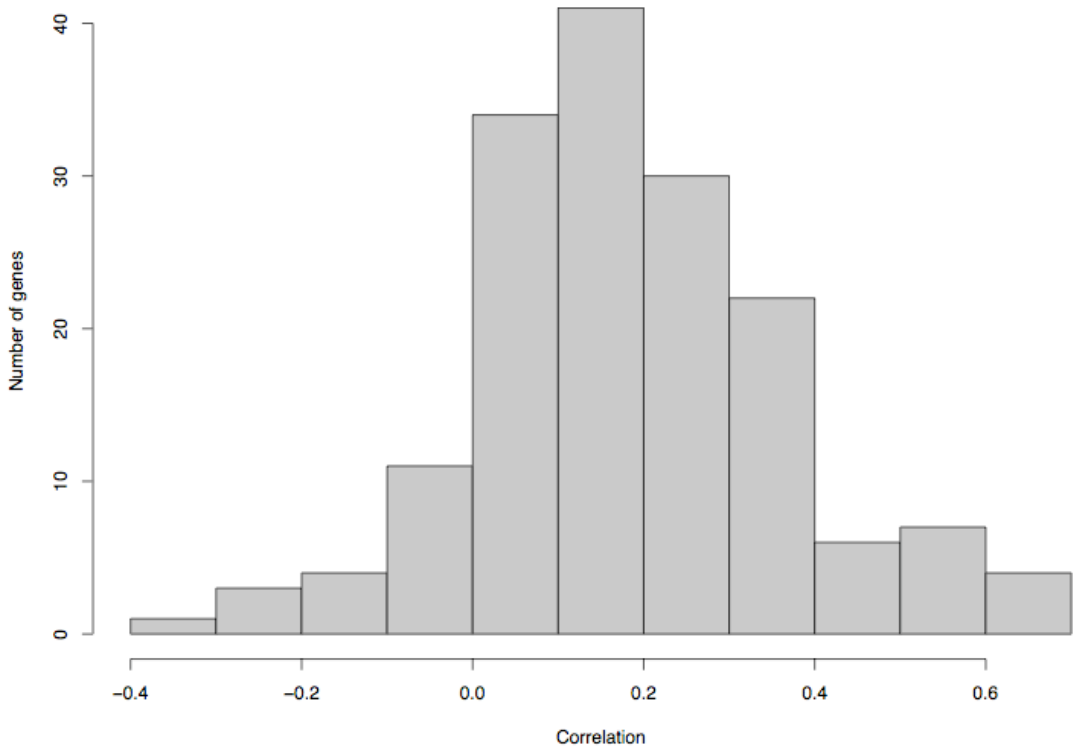


Supplementary Figure 1: Well separated versus scrambled cliques. The illustration on the left depicts an ideal case, in which cliques are well separated. The illustration on the right depicts a more realistic case, in which the cliques are scrambled.



Supplementary Figure 2: Log-log plot of ion current peaks. Two sets of quadruplicate LC-MS/MS analyses were performed on a single segregant from the cross described. Intensities from the first four data sets and from the second four data sets were aggregated. The logrithms of these aggregate intensities are plotted on the x and y axes, respectively.



Supplementary Figure 3. Correlation between transcript levels and protein levels. The graph shows the correlation between relative peptide and transcript abundance for the 163 peptides with a single unique protein match and corresponding transcript data.

peptide	protein	linkage marker	lod score	marker distance from middle of gene
SVVIHAGQDDLGKGDTEESLK	YJR104C	YJR103W_1873;YLR285W_2333;	4.4	1695
NQAAMNPSNTVFDAK	YAL005C	NAL002C_307;gOL02_2894;	5.4	2469
ATVPVLEQQGTVITR	YGR234W	YCL026C_573;gGR12_1350;	3.4	3507.5
IVPAEDGSGAGAAVIAALAQK	YGL253W	gGL01_1127;	4.2	5765
LPNSNVNIEFATR	YNL055C	gNL07_2716;	5.3	6579.5
FQPQDSTTNPSLILAAAK	YLR354C	YLR350W_2376;	8.8	8852.5

Supplementary Table 1. Peptides showing local linkage. 6 peptides showed linkage to a marker that was less than 20 kilobases from the gene encoding that peptide.

Supplementary Note

Polymorphism rates. The average number of SNPs across the set of 278 genes encoding proteins that exactly matched peptides used in the linkage analysis was 3.4, vs. 7.4 in the total set of 6,229 genes.

Correlation between transcript levels and protein levels. We computed the correlation between relative peptide and transcript abundance for the 163 peptides with a single unique protein match and corresponding transcript data. The correlations ranged from +0.64 to –0.3, with an average of 0.186 (**Supplementary Fig. 3**). This distribution is significantly skewed toward positive values, indicating that for some genes, peptides and transcripts are positively correlated, but these correlations are generally weak.

Comparison of differences between parent strains in transcript and protein abundance. 59 proteins differ in abundance between the parents and have transcripts that were different at $P < 0.005$ [1], constituting 43% (59/137) of the proteins that differ between the parents. At random, the expected number is either 33.7 (if we assume 1528 differentially expressed genes out of 6215 on the array) or 36.6 (if we exclude dubious ORFs to get 5727 genes). Either way, 59 is significantly bigger (chi-squared of 13.8, $P = 0.0002$ or 19.0, $P < 0.00002$), so the differentially expressed proteins are more likely to corespond to the differentially expressed transcripts. We expect very few false positives among the remaining 78, both by statistics and by westerns, so these are probably a combination of (1) false negatives in the transcript data, since we expect around half of the transcripts to be different based on FDR calculations, and (2) real cases where

proteins differ and transcripts do not.

Differences between loci that affect transcripts and those that affect proteins. In order to further explore the relationship between genetic regulation of the transcriptome and the proteome, we compared the loci that influence peptide and transcript abundance for the set of 163 genes with peptides that uniquely match a single protein and for which we have corresponding transcript abundance data. 71 of the 163 peptides had at least one linkage with LOD > 3, and there was a total of 93 such distinct linkages. 110 of the 163 transcripts had at least one linkage with LOD > 3, and there was a total of 140 such distinct linkages. Of the 93 distinct loci linking to peptide abundance, 18 also linked to the abundance of the corresponding transcript with LOD > 3, 19 linked to the abundance of the corresponding peptide with 1 < LOD < 3, and 56 did not show linkage to the abundance of the corresponding peptide (LOD < 1). Of the 140 distinct loci linking to transcript abundance, 17 also linked to the abundance of the corresponding peptide with LOD > 3, 31 linked to the abundance of the corresponding transcript with 1 < LOD < 3, and 92 did not show linkage to the abundance of the corresponding transcript (LOD < 1). Thus even with a very relaxed linkage threshold, the majority of loci influenced either peptide abundance or transcript abundance but not both.

In order to examine peptide linkages independant of transcriptional influence we performed linkage of peptide levels with expression of corresponding transcripts treated as covariates. Of the 109 loci detected originally, 79 were also detected in this new analysis, along with 17 additional linkages. This result clearly shows that most of the loci that influence peptide levels do so independently of any effects on transcript levels, and

thus presumably reflect polymorphisms in posttranscriptional regulation. It is worth noting that there was significant overlap in the locations of hotspots for transcripts and proteins: 3 out of the 4 hotspots for protein regulation are reflected at the transcript level, something that could not have happened by chance ($p < 0.0001$). This likely reflects single mutations that have differential effects on the transcriptome and the proteome.

1. Brem, R.B., Yvert, G., Clinton, R. & Kruglyak, L. Genetic dissection of transcriptional regulation in budding yeast. *Science* **296**, 752-5 (2002).

Supplementary Methods

QUANTITATION OF ION CURRENTS

The setting and the notation

The raw LC-MS/MS data files are converted to text format using Thermo Electron's X-Calibur software. A Perl script is then used to generate files that follow the format (h_j, m_j, I_j), where h_j is a scan number (used as a time stamp), m_j is the mass to charge ratio (m/z) (rounded to the nearest Dalton) and I_j is the ion current intensity (used for relative quantification). Data are stored into *Raw pamphlet* , which is a matrix $P_{h,m}$, where *h=1,MaxScan* and *m=1,..,maxM/z*. The matrix entrees are $P_{h_j,m_j} = I_j$

Normalization and data reduction

Mass-intensity bias. Ion current intensity, I_j, often depends on mass m_j. More precisely, if we fix an m/z and add all the intensities from *Raw pamphlet* (i.e. $I_{mz} = \sum_h P_{h,m}$), the two sequences $\{m/z, I_{mz}\}$ will exhibit a linear dependency. In order to compensate for this, we use a simple robust regression line (i.e. $\hat{I}_{mz} = A + Bm$) and normalize all the intensities accordingly.

Data reduction. After Mass-intensity bias is fixed, we collect all the observed intensities and treat them as random variables. The majority of the data constitute random noise and have a Gaussian structure. On the other hand, the desired mass spec peptide peaks behave like outliers and can be observed as a heavy right hand tail (the intensity of peaks are often 1-3 orders of magnitude higher than average noise). Therefore by fitting a Gaussian curve on lower portion of data we can eliminate a large portion of the noisy data.

Outliers. Some peptides ionize extremely well and produce intensities I_j that are several orders of magnitude higher then the majority of peaks. This creates difficulties for filtering algorithms. We developed a simple procedure that deals with these outliers. Starting with the top 5% of the data (i.e. top 5% of $P_{h,m}$'s), we construct an empirical cumulative distribution function $F_n(t)$. Next we fit an exponential curve in such way to minimize $\int (F_n(t) - 1 + e^{A+Bt})^2 dt$ and recompute the top 1% of intensities accordingly. With this approach we smooth out the outliers in a natural way. Although these manipulations alter the data considerably, the relative sizes remain undisturbed.

Signal Filtering

Pamphlet normalization: We have previously (2) described the *M*-*N* rule, which essentially extracts only those pixels that are *M* times higher then median for *N* headers in a row. Since this filtering is self-normalized (the medians are computed for each data set separately), applying the *M*-*N* rule on two different experiments (with fixed *N* and *M*) results in a very similar number of extracted pixels. However, we experimentally observed that +/- 10% of variation in numbers of extracted pixels still remains (for example 5-6 rule in one experiment would produce 22000 pixels while the same 5-6 rule applied to a second experiment would produce 24000 pixels). In order to further normalize the data, we derived an optimization scheme that finds *M*-*N* that extracts a predetermined number of pixels up to 1% precision. Consequently, each data set will have slightly different *M*-*N* rules, but the number of pixels will be within 1% tolerance. Since, by definition, reproducible data sets should have similar M-N rules, we use this quantification to help identify low quality data sets.

The final product at this stage is a collection of pixels $Pix_j = (h_j, m_j, I_j)$ where $j = 1,...L$ for some predetermined number *L*. We used *L=64000*.

Peak detection

Since the majority of peptides elute through time and since we often detect more than one isotope per peptide, one expects that a typical peptide would be represented by several pixels. The "feature extraction algorithm" (2) is used to pull out the "peaks" (a collections of pixels corresponding to a single peptide). The final product at this stage is a collection of peaks $Pk_j = (hc_j, mc_j, V_j)$, where hc_j and mc_j are "header center" and "mass center" respectively. The integrated ion current of all the pixels belonging to the peak Pk_j will be denoted by V_j.

Peptide Sequence Insertion

In a typical LC-MS-based profiling experiment, peptide mixtures derived from protein digests are pumped through a chromatography column packed with reverse-phase beads and electrosprayed into an online tandem MS instrument. The instrument alternatively records the signal intensities (*I*), mass-to-charge (*m/z*) ratios and the scan number (*h*) of all of the detectable eluting peptides, as well as the fragmentation pattern of individual peptides subject to collision-induced dissociation (CID). Database search and comparison of the fragmentation patterns, are then used for peptide sequencing using SEQUEST [5].

As a result we have list of peptide sequences (Pep_j), accompanied with the scan numbers (h_j) and parent mass over charge ratios (m_j). We use a simple linear transformation $\tilde{h}_j = A + Bh_j$, where we chose the parameters *A* and *B* in order to

maximize the overlap between the peaks $Pk_j = (hc_j, mc_j, V_j)$ and peptide sequences $(\tilde{h}_j, m_j, Pep_j)$. As optimization algorithm we used the ARS [6].

Alignment

For two different LC-MS data sets our method produces two sets of peaks $\{Pk_{j,1}\}_{j=1}^{M_1} = (hc_{j,1}, mc_{j,1}, V_{j,1})_{j=1}^{M_1}$, and $\{Pk_{j,2}\}_{j=1}^{M_2} = (hc_{j,2}, mc_{j,2}, V_{j,2})_{j=1}^{M_2}$. The key issue is to compare the total ion currents of these peaks, and since the indexing is arbitrary one cannot just compare $V_{j,1}$ with $V_{j,2}$. The only way to do this is to "align" the time stamps and mass-over-charge ratios (hc's and mc's) and then deduce which intensities to compare (say $V_{j,1}$ with $V_{k,2}$ for some j and k).

Formally, we need to derive a time transformation function Φ, which would maximize the alignment between transformed peaks $\tilde{P}k_{j,2} = (\Phi(hc_{j,2}), mc_{j,2}, V_{j,2})$ and the starting peaks $Pk_{j,1}$. The details of this algorithm and related issues are presented in (2). The transformation function Φ is not linear and consequently finding the optimal Φ is not trivial. Moreover, since we are essentially dealing with image processing, and since the number of alignments increases quadratically with increasing numbers of data sets (for n data sets we need n^2 alignments), this part of our algorithm is by far the most computationally demanding.

The earlier version of this software [7] dealt with several hundred peaks and only 20 different LC-MS data sets. The experiment presented here involves several thousands peaks and 400 LC-MS data sets, which in turns implies a 1000 fold increase in computational complexity (what once took hours now would take months). For this reason we developed the following two modifications:

Adaptive search: Instead of optimizing the alignment with 5 dimensional splines directly, we developed a two-step approach where we first align the pamphlets using a simple linear modification. This can be done very quickly since we have only two variables, and then we use this approximation as a starting point for 5 dimensional spline optimization. This simple trick reduced the number of computations by a factor of ten.

Center of Peaks alignment: Instead of aligning the whole set of 64000 pixels, we work only with the centers of peaks (*hc*'s and *mc*'s). Since there are circa 10 times fewer peaks than pixels this approach speeds up the process by an additional order of magnitude.

Peptide manifolds algorithm

Using the above filtering and peak extraction algorithms we treated the LC-MS data sets and extracted S peaks (peptides) from each of K data sets. The integrated ion current for each of the peak will be denoted by $V_{m,k}$ where *m=1,...,S* and *k=1,...,K*. In our case *K=400* (runs) and $S \approx 3000$ (peptides).

The alignment algorithm above provides us with a pair-wise connections matrix *M*, where $M(V_{m,k}, V_{i,j}) = 1$ will mean that $V_{m,k}$ & $V_{i,j}$ are connected (i.e the two peaks from two different LC-MS data sets have been aligned and thus we believe that they belong to the same peptide or $V_{m,k} \leftrightarrow V_{i,j}$).

Since our main goal is to trace the peptide intensity variations through numerous samples (segregants), the pair-wise comparison is just a start. What we really need is a *Peptide array* $Pa = V_{m_1,1} \rightarrow V_{m_2,2} \rightarrow ... \rightarrow V_{m_K,K}$ which would allow us to relate peptide variations with genetic variation.

In the ideal case, the construction of a *Peptide array* would reduce to a simple, linear application of connections matrix (starting with arbitrary $V_{m_k,k}$ we choose $V_{m_{k+1},k+1}$ so that $M(V_{m_k,k},V_{m_{k+1},k+1})=1$). Unfortunately the alignment procedure is not perfect and it suffers from two major restrictions:

Restriction a) Missing connections: Due to biological and chemical noise, circa 10-40% of the time a peptide found in one LC-MS data set is not present in another data set. Therefore, even if the two vertices $V_{m,k}$ & $V_{i,j}$ do represent the same peptide, the connections matrix will not always indicate the connection (i.e. $M(V_{m,k},V_{i,j})=1$ only 60-90% of times).

Restriction b) Misalignments: Occasionally two peptides with very similar masses elute at very similar times and the alignment algorithm has hard time distinguishing the two. According to our data (See Supplementary Figure 2) in 1-10% of cases the algorithm aligns the peptides incorrectly. (i.e. $M(V_{m,k},V_{i,j})=1$ were it should be equal zero)

Because errors accumulate in exponential fashion, even if we achieve 99% correct alignments, with 400 data sets the probability that the simple approach will work correctly is $0.99^K = 0.99^{400} < 0.02$. In other words even with near perfect pair-wise alignment, the naïve method would almost certainly produce an incorrect array.

In addition, we found that simple ad-hoc methods, the ones that would "self-correct" and add new $V_{m_{k+1},k+1}$ only after comparing it with all the previous $V_{m_1,1},...,V_{m_{k-1},k-1},V_{m_k,k}$ do work reasonable well for moderate size *K* (*K<100*), but tend to encounter problems for larger *K*'s (too sensitive to the choice of a starting point, self conflicting logic, etc.).

We abandoned this local approach of extracting one *Peptide array* at the time, and decided to use graph theory to treat the problem globally. All the LC-MS data sets and their peaks' intensities $\{V_{m,k}\}_{m\le S,k\le K}$ will be treated as vertices of a graph Γ and the connection matrix $M(V_{m,k},V_{i,j})$ will serve as the graph connection matrix.

In the ideal case, the one where we do not have restrictions a) and b) above, *Peptide array* should be easily recognized as a complete sub-graph (Clique). Namely, each Vertex should be connected only to Vertices belonging to the same peptide, thus making the complete graph.

Now it is fairly easy to address the two restrictions. The missing connection would just mean that the sub-graphs in question are not complete but rather "near-complete" while the misalignment issue would imply that these "near-complete" sub-graphs are connected among themselves.

Therefore, from a mathematical point the problem is well defined. All we need is to search for the sub-graphs of the size K, that are a) best connected and b) best separated. An illustration will help: The graph on the left in Supplementary Figure 1 shows an ideal case while the graph on the right shows a more realistic case. We need to produce an algorithm that would recognize most of the scrambled cliques in the realistic case.

Although graph theory provides us with relatively efficient algorithms for extracting fixed size cliques, the sheer size of our graph prevents us from using them. (We have 1.2 million vertices with more than billion connections). We need further simplifications.

Our 400 LC-MS experiments are broken into 100 different strains of yeast, each experiment in quadruplicate. These data sets are very reproducible (circa 90%) and they present a natural starting point.

Step 1: *Extracting all the Cliques of size 4*.

We first run the standard algorithms from graph theory (now admissible since cliques are of very small size) and extract all the cliques of size 4. Since the reproducibility of LC-MS data sets for fixed segregant is 90%, we expect that in 46% of cases (since $1 - 0.9^6 = 46\%$), a clique of size 4 will not be attainable. To address this issue we extract cliques of size 3 and 2 and try to extend them using a "common neighbors criterion".

More formally: For a fixed vertex $V_{m,k}$ we define a neighborhood set

$$Nhd(V_{m,k}) = \{V_{i,j} \in \Gamma \quad \& \quad M(V_{m,k}, V_{i,J}) = 1\}$$

Given a clique $C_3 = \{V_{m_1,1}, V_{m_2,2}, V_{m_3,3}\}$ and index *k*, we find the vertex $V_{X,k}$ that maximizes the neighborhood function

$$F(V_{X,k}) = Card\{\bigcap_{L=1}^{3} Nhd(V_{m_L,L}) \cap Nhd(V_{X,k})\}$$

The clique is now extended to a sub-graph of size 4 depending on the value of the function $F(V_{X,k})$. The rational for this extension is simple: by the very nature of our experiments we expect to see 4 reproducible peaks within the same segregate, and since they will all belong to the same peptide, their connections to the rest of the graph will be similar (i.e. the majority of them will connect only to vertices belonging to the very same peptide).

Step 2: Cliques Connection. We start by defining the cliques connection score function

$$Fc(C_a, C_b) = \sum M(V_{i,j}, V_{k,l}) \text{ where } V_{i,j} \in C_a \text{ and } V_{k,l} \in C_b .$$

Next, we create a "chain of cliques" $C_1, C_2, ...$ by applying the following optimization scheme:

I) Choose C_1, C_2 so that

$$Fc(C_1, C_2) \geq Fc(C_a, C_b) \text{ for all } C_a, C_b \in \Gamma - \{C_1, C_2\}.$$

II) Given $C_1, C_2, ..., C_k$ we choose C_{k+1} so that

$$Fc(C_k, C_{k+1}) \geq Fc(C_a, C_b) \text{ for all } C_a, C_b \in \Gamma - \{C_1, C_2, ..., C_{k-1}\}.$$

We need to emphasize that although a clique is naturally associated with a single yeast strain, the ordering the cliques is not the same as the order of the strains.

Step 3: *Inconclusive versus empty cliques*: Low abundance peptides often manifest themselves as empty (missing) vertices on the graph. However, step 2 above will always produce a sequence of cliques regardless how good (or bad) are the connection scores *Fc*. Thus we need separate set of algorithms that deal with this issue.

The two most common cases are: inconclusive clique (easily recognized since the best and second best scores are very similar, indicating that the algorithm cannot conclusively decide which clique to choose) and empty clique (recognized by the fact that the best score is zero or near zero, indicating that there are no vertices – peptides - present)

The "in between" cases we treated conservatively (i.e. we choose inconclusive status unless we are fairly sure it is a zero clique).

Original Fortran code is available upon request from D.R.

Evaluation of reproducibility and alignment

In order to examine the reproducibility of our method as well as the quality of our alignment algorithm we considered two quadruplicate LC-MS/MS analyses of segregant

1. For each aligned peptide we aggregated 4 intensities (from the quadruplicate LC-MS/MS analyses) and took the logarithm (Supplementary Figure 2).

In the ideal case, this would generate a perfect diagonal line. In practice, we see a deviation that is very close to the perfect case. Two yellow lines, which correspond to +/- 30% deviation of the original data, demonstrate that more than 90% of points are within this range. Moreover this is achieved for thousands of peptides. (It should be noted that, as expected, lower abundant peptides, due to higher "signal-to-noise" ratio, have higher deviation).

In order to achieve such a result, our method needs to overcome two hurdles: First, it must correctly filter out the noise and extract the relevant information from ion peaks; and second, it musst perform all 8 alignments perfectly. Any misalignment or filtering error would result in a pixel far outside the diagonal line.

1. Longtine, M.S. et al. Additional modules for versatile and economical PCR-based gene deletion and modification in Saccharomyces cerevisiae. *Yeast* **14**, 953-61 (1998).
2. Keller, A., Nesvizhskii, A.I., Kolker, E. & Aebersold, R. Empirical statistical model to estimate the accuracy of peptide identifications made by MS/MS and database search. *Anal Chem* **74**, 5383-92 (2002).
3. Brem, R.B., Yvert, G., Clinton, R. & Kruglyak, L. Genetic dissection of transcriptional regulation in budding yeast. *Science* **296**, 752-5 (2002).
4. Brem, R.B. & Kruglyak, L. The landscape of genetic complexity across 5,700 gene expression traits in yeast. *Proc Natl Acad Sci U S A* **102**, 1572-7 (2005).
5. Eng, J.K., McCormack, Ashley L., Yates, John R. III. An Approach to Correlate Tandem Mass Spectral Data of Peptides with Amino Acid Sequences in a Protein Database. *Journal of the American Society for Mass Spectrometry* **5**, 976-989 (1994).
6. Appel, M.J., LaBarre, R.,Radulovic, D. On Accelerated Random Search. *SIAM Journal on Optimization* **14**, 708 - 731 (2003).

7. Radulovic, D. et al. Informatics platform for global proteomic profiling and biomarker discovery using liquid chromatography-tandem mass spectrometry. *Mol Cell Proteomics* **3**, 984-97 (2004).